Spencer Teetaert, PhD Student

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https://scholar.google.com/citations?user=i5a7uuoAAAAJ&hl=en&oi=ao

Research Interests

My research focuses on stochastic dynamic state estimation for continuum and soft robotic systems, with an emphasis on continuous-time probabilistic methods for highly compliant structures operating under uncertainty. I aim to enable robust deployment of continuum robots in real-world, unstructured environments beyond controlled laboratory settings.

Education

2023 – present — **Ph.D. Student in Robotics Engineering, University of Toronto** under the supervision of Dr. Timothy D. Barfoot and Dr. Jessica Burgner-Kahrs. Research focus on stochastic state estimation for continuum robots.

2018 – 2023 — **B.A.Sc. Engineering Science, University of Toronto** in Robotics. GPA: 3.77 / 4.0
Thesis: *Towards Learning-Based Control for a Tendon-Driven Parallel Continuum Robot*.

Honours and Awards

2026-27	**Ontario Graduate Scholarship**
2025-26	**University of Toronto Robotics Institute Fellow**
	Queen Elizabeth II Graduate Scholarship in Science and Technology
2024-25	**Ontario Graduate Scholarship**
2022	**NSERC Undergraduate Student Research Award**
2018	**Faculty of Applied Science & Engineering Admission Scholarship**
	Dr. Dale Iwanoczko Memorial Scholarship

Publications

Peer-Reviewed Journal Articles

1. **S. Teetaert**, W. Zhao, A. Loquercio, S. Zhou, L. Brunke, M. Schuck, W. Hönig, J. Panerati, and A. P. Schoellig, “Advancing reproducibility, benchmarks, and education with remote sim2real: Remote simulation to real robot hardware,” *IEEE Robotics & Automation Magazine*, vol. 32, no. 1, pp. 117–123, 2025. DOI: 10.1109/MRA.2025.3527291.

Conference Papers

1. **S. Teetaert**, G. Caroleo, M. Pontin, S. Lilge, J. Burgner-Kahrs, T. D. Barfoot, and P. Maiolino, “Continuum robot localization using distributed time-of-flight sensors,” in *Proceedings of Robotics: Science and Systems*, Sydney, Australia, Jul. 2026, forthcoming.
2. **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “A sliding-window filter for online continuous-time continuum robot state estimation,” in *2026 IEEE 9th International Conference on Soft Robotics (RoboSoft)*, 2026, pp. 239–246. DOI: 10.1109/RoboSoft67810.2026.11522922.

Preprints

1. A. Gotelli, T. Zheng, A. Peloso, **S. Teetaert**, P. T. Dewi, G. Zetko, C. Shentu, and J. Burgner-Kahrs, *Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion*, 2026. DOI: `10.21203/rs.3.rs-10044366/v1`.
2. **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, *A stochastic framework for continuous-time state estimation of continuum robots*, Conditionally accepted at IEEE Transactions on Robotics, 2025. arXiv: `2510.01381`.
3. **S. Teetaert**, W. Zhao, N. Xinyuan, H. Zahir, H. Leong, M. Hidalgo, G. Puga, T. Lorente, N. Espinosa, J. A. D. Carrasco, K. Zhang, J. Di, T. Jin, X. Li, Y. Zhou, X. Liang, C. Zhang, A. Loquercio, S. Zhou, L. Brunke, M. Greeff, W. Hoenig, J. Panerati, and A. P. Schoellig, *A remote sim2real aerial competition: Fostering reproducibility and solutions' diversity in robotics challenges*, 2023. arXiv: `2308.16743`.

Other (Poster Presentations, Workshops, etc.)

1. **S. Teetaert**, "Noisy noodles: Stochastic state estimation for continuum robots," Toronto Robotics Conference (TRC), 2026.
2. **S. Teetaert** and J. Burgner-Kahrs, "Toward uncertainty-aware intelligence in soft robotics," Workshop on AI-Driven Soft Robotics: Innovations, Challenges, and Future Directions, IEEE International Conference on Robotics and Automation (ICRA), 2026.
3. **S. Teetaert***, E. Walsh*, E. Diller, T. D. Barfoot, and J. Burgner-Kahrs, "Sliding sensors: Configurable confidence in state estimation for continuum robots," IEEE 9th International Conference on Soft Robotics (RoboSoft), 2026.
4. **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, "Space-time continuum: Continuous shape and time state estimation for flexible robots," 40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA40), 2024.
5. **S. Teetaert**, "Accessible sim2real for benchmarking and hands-on education," Workshop on Benchmarking, Reproducibility, and Open-Source Code in Controls, IEEE Conference on Decision and Control (CDC), 2023.
6. Co-organizer, "Safe robot learning competition," `https://www.dynsyslab.org/iros-2022-safe-robot-learning-competition/`, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022.

Employment History

Teaching Experience

2024–present — **ROB498:** Practical teaching assistant for engineering undergraduate Robotics Capstone Design. Head TA as of 2026. Assists in course design and technical aid to students building custom drones.

2023 — **ROB301:** Marking teaching assistant for undergraduate Introduction to Robotics.

Industry Experience

2021–2022 — **Research Engineer Coop,** Human Machine Interaction Lab, Huawei Canada.
Developed advanced machine learning computer vision models for eye-gaze and gesture tracking with applications to consumer electronic devices.

Employment History (continued)

2019–2020 **Industrial Engineering Coop,** Hylife Foods
Launch an automation overhaul project for a production plant of 2,000+ workers. Built relationships with third party robotics partners and aided in design projects. Lead a cutting blade optimization project.

Service

Professional

2026 **Reviewer**, IEEE Robotics and Automation Letters, IEEE Transactions on Robotics

2025 **Reviewer**, IEEE International Conference on Soft Robotics (Robsoft) 2026, IEEE International Conference on Robotics and Automation (ICRA) 2026

2022 **Co-organizer**, Safe Robot Learning Competition, *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*

Extra-Curriculars and Volunteer

2025–2026 **University of Toronto Undergraduate Research Students' Association**
- *Graduate Student Mentor*

2020–2023 **University of Toronto Robotics Association**
- *President*
- *Project Manager, Autonomous Rover Team*
- *Computer Vision Designer, Autonomous Rover Team*

2019–2023 **University of Toronto Engineering Orientation**
- *Subcommittee Chair, Photography*
- *Subcommittee Chair, Retreat*
- *Group Leader*